DarkGuard AI: Product Requirements Document (v3, Hackathon Scope)

Product: DarkGuard AI, an intelligent auditor for deceptive e-commerce design **Type:** Chrome/Edge browser extension with a small public website **Status:** MVP scope locked for the hackathon, with one ML component

---

1. Problem

Some e-commerce sites use dark patterns: interface techniques that push shoppers toward decisions they would not make with full information. Common examples are fake countdown timers, misleading discounts, guilt-laden decline buttons, and add-ons pre-selected at checkout. Most shoppers do not recognize these patterns, and those who do have no easy way to see the page without the pressure.

2. Solution

DarkGuard AI scans the product page the user is viewing, detects a defined set of dark patterns, explains each finding with evidence, and offers a reversible **Ethical View** that annotates suspicious elements.

Detection uses a hybrid design with a clear division of labor:

- **Machine learning proposes.** A small text classifier reads short UI strings (buttons, labels, banners) and estimates which dark-pattern categories they may belong to.
- **Rules verify.** Deterministic checks look for concrete evidence: a frozen timer, an arithmetic mismatch in a discount, a pre-checked add-on box. A finding backed by rule evidence is the primary detection.
- **Approved templates change the page.** Ethical View applies only annotations from a fixed catalogue. No model writes code, CSS, or page content.

DarkGuard never changes prices, form values, or purchase controls, and never submits or completes anything on the user's behalf.

3. Goals for the hackathon

1. Reliably detect four dark-pattern categories on a single product page.
2. Use ML for text-based detection and show, with measurements, that it improves on rules alone.
3. Show evidence and a plain-language explanation for every finding.
4. Highlight the exact element on the page.
5. Produce a transparent 0–100 risk score.
6. Demonstrate a reversible Ethical View with a one-click "Show original" toggle.
7. Measure everything on a labeled test set rather than claim accuracy.

Non-goals are listed in section 5.2.

4. Target users

User	Need
Online shoppers	Understand pressure tactics on the page in front of them
Web developers and designers	See which of their UI patterns could be read as deceptive
Researchers and students	A reproducible tool for studying dark patterns

Website-owner audit reports are a later feature.

5. Scope

5.1 In scope (MVP)

- Chrome and Edge extension (Manifest V3).
- Detection of four categories (section 6), using rules and the text classifier together.
- Local detection that runs in the browser, with no page data sent anywhere by default.
- A baseline text classifier (TF-IDF plus logistic regression) trained offline and shipped as a small JSON file.
- Template-based explanations, with optional AI-written explanation text when a backend is available.
- Element highlighting and a 0–100 risk score.
- Ethical View: annotations, a banner, and a "Show original" toggle.
- Local scan history (last 20 scans).
- Landing page with install instructions, demo video, and a link to the test site.
- A mock e-commerce test site with planted and clean pages, used for demo and evaluation.
- An evaluation report comparing rules-only against rules plus ML.

5.2 Out of scope (later versions)

- Transformer or embedding-based text models.
- Generative models that write or change page code (rejected by design, see section 8).
- Hidden-cost and sneak-into-basket detection across multiple checkout pages.
- Subscription trap detection (requires multi-step flows).
- Computer vision and screenshot analysis.
- User accounts, database storage, and cloud history.
- Firefox and Safari support.
- Website-owner analytics and enterprise audits.

6. Dark patterns in scope

Each category has a rule-based evidence check. Where text matters, the ML classifier contributes a probability.

ID	Category	Rule evidence (single page)	ML contribution	Default severity
DP-1	**Fake urgency**	A countdown or deadline stays unchanged for 10 seconds.	Classifies urgency wording ("only 2 left", "hurry") when no timer is present.	High
DP-2	**Misleading discount**	Reference price and stated percentage differ from the calculated one by more than 5 points.	Classifies discount wording when the reference price is unlabeled.	Medium
DP-3	**Confirm shaming**	Decline option has guilt-based wording near the primary action.	Main detector for this category: classifies decline-option text.	Medium
DP-4	**Pre-selected extras**	Add-on checkbox (insurance, warranty, gift wrap, donation, subscription) is checked by default near the add-to-cart control.	Classifies the checkbox label as an add-on when keywords miss it.	Medium-High

Design rule: A finding is presented as a *signal* that a pattern may be present. It is never presented as a legal finding or proof of intent.

Evidence rule: A detection shown with a strong label must have rule evidence. A classifier-only detection is shown as a lower-certainty signal (section 7, FR-13).

7. Functional requirements

ID	Requirement	Priority
FR-01	User can install the extension from the landing page or as an unpacked developer build.	Must
FR-02	Extension requests access to the current site only when the user clicks "Scan this page."	Must
FR-03	Extension detects DP-1 to DP-4 using the rules and classifier in section 6.	Must
FR-04	Each finding shows category, evidence, confidence, severity, and explanation.	Must
FR-05	Each finding highlights its element and scrolls to it on click.	Must
FR-06	A 0–100 risk score is shown with its level.	Must
FR-07	Ethical View applies annotations from the approved catalogue only.	Must
FR-08	"Show original" removes all Ethical View changes and restores the page exactly.	Must
FR-09	Local scan history (last 20 scans) can be viewed and cleared.	Should
FR-10	User can export a scan as JSON or Markdown.	Should
FR-11	Optional backend rewrites explanation text. The extension works without it.	Should
FR-12	Landing page explains the product and links to the install guide and test site.	Must
FR-13	Classifier-only findings (no rule evidence) are shown as "possible pattern, low certainty," capped at confidence 0.5, and use caption templates only.	Must
FR-14	If the classifier fails to load or times out, the scan completes with rules alone and the UI says so.	Must
FR-15	The scan result records which model version produced each ML contribution.	Should

8. Machine learning requirements

ML is used for text classification only. The following rules hold for every model version.

ID	Requirement	Priority
ML-R1	The classifier runs locally in the content script, with no network call at inference time.	Must
ML-R2	One inference takes under 5 ms on a typical product page (about 200 candidate strings).	Must
ML-R3	The model is a versioned JSON artifact (`shaming-v1.json` naming pattern) with a SHA-256 checksum verified on load.	Must
ML-R4	The classifier outputs probabilities and category labels only. It cannot create templates, change page content, or touch form controls.	Must
ML-R5	Correction selection is a lookup from category to approved template IDs. No model is used to choose or write corrections.	Must
ML-R6	Generative models are not used to create or modify page content, CSS, or scripts.	Must
ML-R7	A model version is enabled by default only if it meets the release gate in ML-R9.	Must
ML-R8	Training data contains only short UI strings and their labels. No personal data, form values, or full page HTML. Copied third-party data is used only under its license, and its source is recorded.	Must
ML-R9	Release gate: precision ≥ 0.85 on the held-out clean set, recall ≥ 0.70 on the planted set, and no increase in false-positive rate over rules-only on clean pages.	Must
ML-R10	Each model version ships with a model card (training data, split method, metrics per category, known weaknesses).	Must
ML-R11	Training code, labeling guidelines, and splits are committed to `ml/` so results can be reproduced.	Must

9. Ethical View behavior (product guarantees)

These guarantees hold regardless of how a detection was produced. They are enforced by tests.

1. **No price changes.** Ethical View never edits, hides, or reformats a price.
2. **No form changes.** It never changes checkbox state, input values, or selections.
3. **No blocked controls.** Buy, add-to-cart, and checkout controls stay visible and clickable.

4. **Labeled, not deleted.** Suspicious elements get a visible note.
5. **Reversible.** One click restores the original page.
6. **Neutral wording.** Annotations describe what was observed and avoid telling the user what to do.

10. Non-functional requirements

Area	Requirement
Performance	Local scan completes in under 1 second for rules and classifier, excluding the 10-second timer observation window. Optional AI explanations add at most 3 seconds and never block local results.
Privacy	Nothing leaves the browser by default. If the backend is used, only detection summaries are sent, never page HTML, form values, or personal data.
Reliability	If the extension or model fails, the page works normally. Ethical View errors roll back automatically.
Accessibility	Annotations meet WCAG AA contrast and are readable by screen readers. Toggle controls are keyboard reachable.
Compatibility	Chrome and Edge, current stable versions.

11. Success metrics

Measured on the labeled test set (TRD section 12). The set contains at least 30 pages: 20 with planted patterns and 10 clean pages, plus a held-out set of UI strings for the classifier.

Metric	Target
Precision per category (rules plus ML)	0.80 or higher
Recall per category on planted pages	0.75 or higher
False-positive rate on clean pages	10% or lower
ML contribution (ablation)	Recall improves over rules-only without raising false positives
Classifier precision (DP-3 text)	0.85 or higher
Local scan time (median, excluding DP-1 window)	Under 1 second
Ethical View restore success	100%
Ethical View guarantee violations	0

Report honest numbers, including what the tool misses and which categories are weak.

12. Risks and mitigations

Risk	Mitigation
Classifier overfits to the training sites	Split by site, not by string. Report per-site results. Keep the rule evidence requirement.
Too little labeled data	Start with 600–1,000 short strings from the test site and hand-written examples. Use a licensed public dataset only if its terms allow it.
False accusations against real businesses	Describe findings as signals. Show evidence. Give the user the option to view the original page.
Model file is tampered with or fails to load	SHA-256 check on load. Fall back to rules alone and tell the user.
Ethical View breaks a site's layout	Use additive annotations only. Test on the test site and two real sites.
Live demo fails due to network	Detection works offline. Pre-record a backup video.
Scope creep during the event	Freeze the four categories by the end of day 3. New ideas go to a "later" list.

13. Milestones (7 days)

Day	Deliverable
1	Test site with planted and clean pages. Extension skeleton with popup and scan button. Labeling guidelines written.
2	DP-1 and DP-4 rule detectors with fixture tests. Labeled string set started.
3	DP-2 and DP-3 rule detectors. Risk score. Feature freeze for detection categories.
4	Train TF-IDF plus logistic regression baseline. Export JSON, run parity tests, measure latency.
5	Ablation: rules-only versus rules plus ML on the test set. Enable ML only if the release gate passes. Highlighting, explanations, history.
6	Ethical View with full catalogue and "Show original." Evaluation report. Landing page.
7	Demo script, backup video, model card, documentation. Freeze features.